

LEONARDO CHIMAL

309-278-3618 | leochi2565@gmail.com | 755 20th Avenue, East Moline Illinois
<https://leonardochimal-portfolio-production.up.railway.app> | [linkedin.com/in/leonardo-chimal-a442b6267](https://www.linkedin.com/in/leonardo-chimal-a442b6267)

PROFILE

Engineering student with experience in CAD-driven prototyping, robotics, mechanical design, and software development. Skilled in combining mechanical systems, embedded electronics, and software to create functional prototypes. Seeking an engineering internship where I can contribute to product design, prototyping, and real-world system development.

EDUCATION

Black Hawk College | Moline, Illinois | May 2027
GPA | 3.36

United Township High School | East Moline, Illinois | May 2025
GPA | 4.1 (weighted)

EXPERIENCE

John Deere | Moline, Illinois | June 2024 – Current Apprentice IT

- Developed internal tools and automation workflows using AWS, Terraform, JavaScript, and API integrations to support scalable backend systems and reduce manual processes
- Designed and deployed a secure URL shortening service with metadata tracking, encoding/decoding logic, and optimized backend performance for internal production use
- Built and integrated cloud-based services (AWS Lambda, SNS, SQS) using infrastructure-as-code to support event-driven system architecture and workflows
- Mentored incoming IT apprentices by sharing onboarding knowledge, technical guidance, and best practices for internal systems and workflows.

First Tech Challenge | Team 8813 | May 2022 – 2025 Robotics Design Lead

- Led CAD design and mechanical integration for competitive FTC robotics systems using Onshape, collaborating with programming and manufacturing teams to optimize robot performance, maintainability, and weight distribution through iterative prototyping and engineering reviews.
- Designed drivetrain, intake, launcher, and structural robot systems while working alongside engineers and mentors from John Deere, Lockheed Martin, Meta, Tesla, Amazon, Black Hawk College, and other STEM organizations.
- Helped implement Agile/Scrum-style workflows using Trello, Discord task tracking, and sprint reviews while contributing to STEM outreach, mentoring, engineering presentations, and robotics demonstrations throughout the Quad Cities region.
- Qualified for the FTC World Championship in Houston, Texas (2024), FTC State Championship competitions (2025), and the FTC World Championship in Indianapolis, Indiana (2026).

PROJECTS

Autonomous AI Drone Platform

URL: <https://leonardochimal-portfolio-production.up.railway.app/projects/drone>

- Designed and prototyped a modular 5-inch autonomous drone platform in Onshape integrating embedded electronics, power distribution, vibration-isolated mounting systems, and future autonomous system integration. Focused on iterative CAD development, component integration, weight optimization, and maintainable mechanical design workflows.

Languages: JavaScript, Python, SQL

CAD/Engineering: Onshape, CAD, 3D Printing, Drone Systems, Embedded

Development: React, Next.js, APIs, AWS

Meta Quest Dual Controller Slide-Lock Attachment

URL: <https://leonardochimal-portfolio-production.up.railway.app/projects/meta-quest>

- Designed and prototyped a custom Meta Quest dual-controller slide-lock attachment focused on improving two-handed stability, ergonomic handling, and mechanical locking functionality through iterative CAD development and physical testing. Developed multiple design iterations using Onshape and PETG 3D printing workflows to refine usability, structural durability, and controller alignment accuracy.

Languages: JavaScript, Python, SQL

CAD/Engineering: Onshape, CAD, 3D Printing

Prototyping: Mechanical Design, Product Design, Ergonomics